

UCS645

Parallel & Distributed Computing

Assignment Report

Name: Rohini Bansal

Roll Number: 102317175

Topic: Correlation Computation using OpenMP

1. Objective

To compute the correlation coefficient between every pair of input vectors using both sequential and OpenMP-based parallel implementations, and to analyze performance improvements using multithreading.

2. Problem Description

Given n_y input vectors (rows) each containing n_x elements, stored in row-major format, the task is to compute the correlation coefficient for all $0 \leq j \leq i < n_y$ and store the results appropriately in an output matrix.

3. Methodology

The sequential implementation computes mean, standard deviation, and correlation for each pair of vectors. The optimized parallel version precomputes means and norms and uses OpenMP to parallelize independent computations across multiple threads.

4. Performance Results (Matrix Size: 2000 x 2000)

Threads	Sequential Time (ms)	Parallel Time (ms)	Speedup
1	6929	9465	0.73x
2	9297	7242	1.28x
4	7486	4321	1.73x
8	9868	2725	3.62x

5. Observations

- 1 Parallel execution improves performance as thread count increases.
- 2 Initial slower parallel performance is due to thread overhead.
- 3 Maximum observed speedup was 3.62x at higher thread counts.
- 4 Speedup is limited by memory bandwidth and synchronization overhead.

6. Conclusion

The OpenMP-based parallel implementation significantly reduced execution time compared to the sequential baseline. Optimization techniques such as precomputing means and norms improved efficiency and scalability.